

TRACE

An Input-Only Wearable Memory

Architecture, rationale, and the case for a memory
that keeps words instead of footage

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The architecture of record is a hand-drawn napkin, photographed 2026-07-18,
transcribed verbatim in Appendix A. On any conflict, the napkin outranks this document.

Abstract

TRACE is a wearable memory system built on one uncompromising invariant: raw media never persists and never leaves the device; only typed text may cross the boundary. On-device perception converts the live world into coordinate-marked, provenance-bearing text at the moment of capture; the pixels and audio that produced it are discarded within milliseconds. A consolidation process — always running in the background, but better nightly — binds the day’s observations into living nodes by spatial and temporal co-occurrence, derives what was never directly observed, and discards failed derivations without ever touching the raw record beneath them. A supplier answers unconstrained questions by handing an ordinary local language model *just enough context* — three memories, not three thousand — every clause of every answer walking back to the exact reading that justifies it, and refusing honestly where no reading exists.

This document is the complete argument for that architecture: its first principles (Chapters 1–2), its forty years of prior art (Chapter 3), the seven invariants that govern implementation (Chapter 4), the pipeline from photons to answers with one lived moment followed through every stage (Chapters 5–6), the machine itself (Chapters 7–11), and — separated with care, because the distinction is the difference between a claim and a hope — *what runs today, what runs on synthetic fakes, and what is not yet built* (Chapters 12–17). The central hypothesis is stated as what it is: unvalidated, falsifiable, and instrumented. Every architectural decision in this document exists to test it faster.

What is real today

The reader deciding whether to join this project should not have to excavate the truth from an argument. This table is transcribed from the build ledgers of the two repositories on 2026-07-21. Each row names its evidence. Staleness warning: these are facts about a fast-moving codebase, correct on the date above and demoted to hearsay a month later.

Component	Status	Evidence
Domain model · claim log · rendering	IMPLEMENTED	t racemem M0–M1, green tests
Binder — reconcile, tombstones	IMPLEMENTED	t racemem M3
Node creation & night shift — guardrails, selector, judge, pruner, remap	ON FAKES	t racemem M4 — implemented against synthetic inputs
Information supplier — cue, retrieve, pack, ground	IMPLEMENTED	t racemem M5
.trace container — format, keyring, sync, boundary	IMPLEMENTED / STUBBED	M6; libsodium crypto adapter stubbed
Extraction cascade	ON FAKES / STUBBED	M7 — logic real, hardware adapters (OCR/ASR/eyes/tap) stubbed
Crown test: shuffled synthetic week → episodes → entities → nightly patterns → cited answer → honest abstention	IMPLEMENTED	tests/test_crown.py, passing
Unit wall	IMPLEMENTED	274 green · 1 xfail (the TODO ledger, each naming its spec section)
First measured extraction gap	IMPLEMENTED	L1-verbatim recall 0.974 (gap 0.026) over 111 real frames; ceilings audited
Falsification gate on real captured weeks	NOT YET	the REAL gate needs real captured weeks — the next milestone that matters
Legacy stack (v0): live capture, hub, nightly job, phone app	IMPLEMENTED	running since June 2026; being strangled, not extended

The number that matters most is one that is absent: frames kept — zero, by construction, in both stacks. There is no code path that writes a frame to disk.

How to read this document

This document is written for six different kinds of reader. Do not read it linearly unless you want to.

- **In 10 minutes:** the Abstract, the table above, and Chapter 6 — one moment followed end to end.
- **ML / CV / data science:** Chapters 5, 9, and 13 — the pipeline, the perception economics, and the evaluation instrument.
- **Cognitive science:** Chapters 2 and 6 — the biological arguments are load-bearing, not decorative, and you are the reader best placed to break them.
- **Backend / systems:** Chapters 7, 8, and 14 — the domain model, the contract-first architecture, and the two-stack strangling plan.
- **Product / frontend:** Chapters 1, 6, and 11 — the problem, the lived example, and what answering actually feels like.
- **Investors and diligence:** the table above, then Chapters 15–17 — honest risks stated against ourselves, position, and the road.

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Part I · Foundations

CHAPTER 1

Introduction and problem statement

1.1 The problem: lives are forgotten by the people who live them

People forget most of their lives. Where the keys were left. What the doctor actually said, as opposed to what anxiety later reconstructed. Whether the door was locked. The name attached to a face met once at a dinner. What the contractor promised, verbally, in the hallway. The moments that matter are sparse, unpredictable in advance, and irretrievable by the time they are needed. Human episodic memory was never designed for retrieval on demand; it is a reconstructive process, optimized for gist over fidelity, and it degrades precisely along the dimensions — exact words, exact places, exact times — that everyday questions require [1][2].

The obvious remedy is to record everything. It fails twice, and both failures are fundamental rather than incidental.

The first failure is social and legal. A person wearing an always-on camera that stores video is a walking surveillance device. Every bystander is recorded without consent; every stored hour is subject to subpoena, theft, breach, and misuse; every intimate space the wearer enters becomes an archive. The history of wearable cameras is the history of this failure: from the earliest sousveillance experiments to the public rejection of camera glasses, storage-first capture has been socially unacceptable everywhere it has been tried at scale [10][34]. No privacy policy fixes this, because policies are promises, and promises about stored media are exactly what nobody believes.

The second failure is functional. Raw footage is the wrong artifact. Nobody rewatches their life; a day of video is a day long. What people actually want from a memory is not playback but answers: to query the past the way one queries a database, in natural language, and get back a short, true, cited response. The valuable artifact of a lived day was never the footage — it was the bound, queryable meaning a mind would have extracted from it. Storage-first systems defer that extraction forever, and so never deliver it.

1.2 The problem statement

Build a system that perceives a person’s day through always-on, cheap senses; converts everything to words the instant it happens — never keeping one frame of video or one second of audio; consolidates those words, continuously and best nightly, into bound understanding; and later answers arbitrary questions from that understanding with citations, or refuses honestly.

Three phrases in that statement carry the whole design. “*The instant it happens*” is the privacy architecture (Chapter 12): ephemerality at the source, not deletion after the fact. “*Bound understanding*” is the binder and the node layer (Chapters 5, 10): isolated observations are raw material, not memory — the napkin is explicit that the binder’s output “is not a complete Node, it’s raw material.” “*Or refuses honestly*” is the epistemology (Chapters 11, 13): a memory that fabricates is worse than no memory at all, so refusal is a first-class outcome and fabrication is priced into the headline metric at the cost of a correct answer.

1.3 The central hypothesis

Everything rests on one hypothesis: *perception can be decomposed into channels which reconstruct lossless-enough textual context for a tractable local reasoner to answer arbitrary retroactive questions at near-zero hallucination*. It is unvalidated. It is falsifiable. It is instrumented (Chapter 13), and the falsification gate that would kill or confirm it — scoring the system against real captured weeks rather than synthetic ones — is the project’s named next milestone. Every architectural decision in this document exists to reach that gate faster.

1.4 What TRACE is not

TRACE is not a lifelogger: it keeps no log of life, only text extracted from it. It is not a surveillance device: the artifact surveillance requires is never created. It is not a retrieval-augmented chatbot: retrieval here feeds a memory that was *built for* retrieval, not a pile of documents that happen to exist. It is not a model company: the language model is the last and least special box in the machine, ordinary and swappable by design. And it is not finished: the table on page 3 is the honest boundary between the built and the intended.

1.5 Contributions of this document

(1) A first-principles derivation of an input-only memory architecture from the biology of human memory and the failure history of lifelogging. (2) Seven locked invariants, each with rationale, enforcement, and the failure it forecloses. (3) A complete pipeline exposition, from photons to a cited answer, with one moment walked through every stage. (4) An honest, mechanically-diagnosable evaluation method. (5) A candid account of the project's two stacks — the running v0 and the greenfield rebuild — and the strangling plan between them.

CHAPTER 2

Conceptual foundations

2.1 The standard model: a brain never touches the world

The napkin’s first drawing is not of the product; it is of every mind that has ever existed: [World] → Eyes/Ears → [Brain], with the loop closing back into the world. The annotation beside it is the design’s first principle: *“because we can’t manufacture other senses, or most of the information is obtained through this.”* A brain never touches the world directly. It receives two narrow, lossy, sensory streams, extracts meaning at the moment of perception, and stores the meaning — never the stimulus. TRACE copies exactly this: video for eyes, audio for ears, extraction at the moment of capture, retention of meaning only. The architecture is not biomimicry for its own sake; it is the observation that the only working memory system we know of is input-only, and that its constraints are load-bearing.

2.2 Episodic memory is reconstruction, not playback

Human recall does not replay footage, because there is none. Remembering is reconstruction from consolidated traces, schema-driven and confident even when wrong [2]. Tulving’s episodic/semantic distinction [1] maps directly onto TRACE’s two stores: the time-stamped claim log is episodic — events located in time and place, sparse and literal — while the bound node graph is semantic: consolidated, cross-referenced, queryable. The mapping is load-bearing, not decorative. It predicts the interface between the stores: consolidation runs episodic→semantic, never the reverse, exactly as the night shift reads the log and never writes it. And where human reconstruction silently invents congruent detail, TRACE’s reconstruction is forbidden to assert anything that does not trace to a logged claim — it says “I don’t know” where a human would confabulate. The machine is honest about reconstruction precisely because its biological original is not.

2.3 Sleep consolidation: why the binder runs at night

The decision to concentrate deep binding in a nightly pass is likewise borrowed from biology. Systems-consolidation research shows the mammalian brain replays the day’s hippocampal traces during slow-wave sleep, gradually binding them into neocortical structure [3] — an architecture forced by the same constraint TRACE faces: deep integration is expensive, and the organism cannot afford it while also perceiving. Night is when the device is charging, thermally unconstrained, and free to run the big model over the whole day at leisure. The napkin’s phrase is exact: node creation is *“always running in the background, but better nightly.”* Incremental binding accelerates; the night shift perfects. Hebb supplies the mechanism transposed to text [35]: claims that co-occur in time and place, repeatedly, get bound into shared structure.

2.4 Event segmentation: why capture is gated, not uniform

Human perception does not sample the world uniformly; it carves experience into events at boundaries of change and encodes richly at those boundaries [4]. TRACE’s capture follows the same economics: the cheap channels run continuously precisely because they are cheap, while the expensive channel — deep visual description by a vision-language model — fires only on attention, dwell, or change. Uniform deep capture would be both unaffordable and pointless: most moments do not matter, and the value of a memory is concentrated in the few that do. The scheduler that makes this call is not an optimization detail; it is the system’s model of what deserves to be remembered well (Chapter 9).

2.5 Text as the retention format: the four arguments

The load-bearing decision — keep words, not media — rests on four arguments. *Privacy*: text is the only format whose retention can be made socially and legally safe by construction; a sentence about a bystander is a fundamentally different object from their stored face (the residue this does *not* discharge is treated honestly in Chapter 12). *Sufficiency*: the hypothesis under test is precisely that words carry enough; language is the interlingua in which modern machine reasoning is strongest [7], and retrieval-then-reason over text is the best-understood recall architecture we have [12]. *Economy*: text is thousands of times smaller than the media it replaces; a lifetime fits on a phone. *Auditability*: a user can read every single thing the system knows about them, line by line, and delete none of it while trusting all of it — an impossible offer for embeddings or media.

2.6 The resolution of the central tension

“Remember everything” and “respect everyone” are only in tension while the retained artifact is media. Retain extracted meaning instead and the tension dissolves: recall quality becomes a function of extraction quality — which is measurable and improvable — while the surveillance artifact simply never exists. That is the whole company in two sentences.

CHAPTER 3

Prior art and position

3.1 The memex lineage: total capture as an old dream

Bush’s 1945 memex [36] founded the genre and set its sixty-year trajectory: capture everything, organize later. The lesson of the lineage is that “later” never comes — capture without retrieval-of-meaning is a landfill.

3.2 MyLifeBits and SenseCam: what storage-first taught

Gemmell, Bell and Lueder proved the landfill empirically at life scale [9]: MyLifeBits’ bottleneck was never storage but structure. SenseCam [10] proved the opposite pole — even passive photo capture measurably aids real, clinical memory — while every deployment surfaced the social cost of visible storage. Hoyle et al. quantified that cost from the wearer’s own side: lifeloggers themselves censor, delete, and manage bystander exposure [34]. Gurrin, Smeaton and Doherty’s survey converges on cues-and-facts over archives [37]. Together these justify TRACE’s inversion: total capture, linguistic retention.

3.3 Ego4D: the demand curve, measured

Ego4D [11] defines the benchmark tasks TRACE’s question batteries descend from — and establishes that long-horizon egocentric retrieval is hard *even with the video retained*. That is the honest context for every recall number in this document: the ceiling is not 100%, even for an oracle with footage.

3.4 Retrieval-augmented generation: the supplier’s ancestry

Lewis et al. [12] define the retrieve-then-generate shape the information supplier inherits. TRACE departs from RAG in what is retrieved from: not a corpus that happens to exist, but a memory *constructed for* retrieval, with provenance and confidence as type obligations rather than metadata conventions. Attribution — answers that cite — is promoted from evaluation criterion [31] to construction rule; selective prediction [13] and calibration [20] supply the formal frame for refusal; the hallucination taxonomy of Ji et al. [32] is operationalized as the grounding gate’s checklist.

3.5 The commercial field: an empty quadrant

Plot the field on two axes — physical-world context versus digital-only, and zero retention versus raw-data retained — and the quadrant TRACE occupies is empty. Camera glasses and AI pins retain media; PKM tools and assistant platforms never touch the physical world. The two purpose-built recorders of this generation died at the privacy wall (Chapter 16). Physical context has always required retention; an architecture that supplies context without retention is the first way into that quadrant.

3.6 Knowledge graphs and open vocabulary

The node layer is a knowledge graph in the loose sense of Hogan et al. [14], with one deliberate deviation: open vocabulary everywhere. There is no fixed ontology of entity types or predicates; the world names itself through the extractors’ text, and structure emerges from co-occurrence, not from a schema. The cost of that choice is weaker global consistency; the payment comes at answer time, where evidence is quoted rather than inferred across long relational chains.

CHAPTER 4

The seven invariants

Architecture is what remains constant while everything else iterates. TRACE locks seven invariants. Each is stated with its rationale, its enforcement, and the failure mode it forecloses. Code that violates an invariant does not merge; several are enforced by construction, so violating them does not even compile against the domain model.

4.1 I1 — No raw persistence

Statement. Raw pixels and audio are discarded the instant words are extracted from them. There is no record path, no cache, no “temporary” buffer that survives the perceptual moment.

Rationale. Every stored frame is a liability with no compensating asset: it cannot be queried directly, and it is the artifact whose existence makes the system a surveillance device. Ephemerality at the source is the only privacy claim that survives an adversarial audit — deletion policies, retention windows, and encryption of stored media all reduce to promises.

Enforcement. In `tracemem` the law is inherited from the blueprint in its hardest form: *raw is immutable and no update path exists; pixels never leave the device*. Capture converts and drops in one motion. Honesty note: gold frames used as evaluation oracles exist on the founder’s development machine only — they are the measuring instrument, not the product path, and the phone build has never persisted a frame.

4.2 I2 — Text-only egress

Statement. The only payload type permitted to cross the device boundary is typed text carrying the identifiers of the source events it derives from.

Rationale. Local models may hit a quality wall (Chapter 15); the architecture must allow routing *reasoning* to a stronger model without ever routing *media* anywhere. Typing the boundary makes the safe path the only path.

Enforcement. In the v0 stack this is an eleven-line guard that raises a privacy violation for any non-text payload — short enough to audit in one breath, which is the point. In `tracemem` the boundary lives in the container’s boundary module, with the same contract.

4.3 I3 — Append-only truth

Statement. The claim log is immutable and append-only; it is the single source of truth. Everything derived from it — episodes, entities, patterns, every index — is a disposable, replayable projection.

Rationale. Consolidation is experimental and will be wrong often. If derived structure were the truth, every experiment would risk memory loss. With an immutable log, any projection can be discarded and rebuilt — the napkin’s discarder rule verbatim: *“the marinated content that’s not useful is discarded, NOT the raw material.”*

Enforcement. The store exposes no update and no delete; corrections are expressed as new claims and tombstones (Chapter 7), never as mutation. Replays are deterministic, which the evaluation harness depends on.

4.4 I4 — Provenance everywhere

Statement. Every claim, binding, derivation, citation, and egress payload carries the identifiers of what it derives from. An unsourced assertion is a construction error.

Rationale. Citations are the product; diagnosis is the method (a wrong answer must be traceable to the claim that misled it); audit is the promise.

Enforcement. Constructors reject empty evidence. Provenance is not a logging convention — it is a type obligation, in the value-object discipline of domain-driven design [19]: invalid memory is unrepresentable.

4.5 I5 — Calibrated confidence; refusal over fabrication

Statement. Every derived assertion carries a grade; the grade degrades with derivation depth; links below the floor are refused, not hardened; at answer time, insufficient evidence yields “I don’t know.”

Rationale. A memory’s worth is its trustworthiness. One confident fabrication — “you locked the door” when you did not — costs more trust than a hundred honest refusals. The metric agrees: the scoring instrument subtracts fabrications from corrects, so the optimizer and the ethics point the same way. Modern networks are miscalibrated by default and calibration must be measured, not assumed [20].

Enforcement. Grade is a domain type; *grade degrades with derivation depth* is one of `tracemem`’s hard laws; refusal is the default disposition of weak evidence in the night shift’s judge; abstention is a first-class answer in the crown test.

4.6 I6 — Unconstrained questions

Statement. There is no fixed question taxonomy, no menu of supported query types. Users ask anything, natively.

Rationale. A constrained question surface is a product decision masquerading as a safety decision: it hides hallucination by refusing to hear the questions that would expose it. The honest path is open vocabulary end to end, with calibrated refusal as the safety mechanism instead of a menu.

Enforcement. The query type is a bare text string by design, and the founder’s decision to keep it so is recorded.

4.7 I7 — Nothing is built until a missed question demands it

Statement. Every helper, sensor, channel, and feature must be authorized by a diagnosed failure on a real question from a real capture.

Rationale. Perception systems invite infinite plausible work, and intuition about which channel matters is reliably wrong. The project’s own history supplies the cautionary tale: a 3D world renderer was built on intuition, answered nothing, and was killed. The discovery loop — ask, miss, diagnose, build exactly the missing channel — replaces guessing with evidence.

Enforcement. Governance: a build starts by naming the missed question it answers. The miss-diagnosis log (Chapter 13) supplies those questions mechanically. Even the two modules in `tracemem` that exceed the napkin (a prediction spine and the falsification harness) are explicitly flagged *beyond-napkin* in the constitution file, awaiting a founder ruling — the invariant applied to the repository itself.

4.8 The invariants as a system

The seven interlock. I1 and I2 make the privacy claim structural; I3 and I4 make memory auditable and experimentation safe; I5 and I6 together make “ask anything” compatible with “near-zero hallucination” — the refusal gate absorbs what open vocabulary exposes; and I7 aims the whole machine at evidence. Remove any one and the others weaken: without I3, diagnosis loses its replay; without I4, refusal loses its justification; without I7, the roadmap reverts to guessing.

Part II · The Machine

CHAPTER 5

The pipeline: from photons to answers

This chapter is the napkin’s EXECUTION line, expanded stage by stage. The interactive walkthrough (interactive/architecture-live.html) renders it live; this is its paper form. The pipeline in one line:

[World] → Extraction ■ → TEXT → Binder → Node Creation ■ → Supplier ■ → LLM

5.1 The world: two channels, no recording

Two streams open onto every second: video for eyes, audio for ears — the napkin’s justification annotated on the drawing itself: we cannot manufacture other senses, and most information arrives through these two. Neither stream is a recording. Frames and samples exist transiently in memory, are read, and die; what survives is only what was read from them.

5.2 Extraction: the first black box

Extraction turns raw light and sound into addressed text. Inside the box, four mechanisms:

The blur gate. Every frame is triaged before anything else sees it. A frame whose content cannot be read — motion smear, defocus — is discarded immediately: “*blurry frames discarded*” is on the napkin. A frame that cannot be read is never written; there is nothing to store, leak, or subpoena.

The spend dial. Resolution and framerate follow the world, not a clock (Chapter 9). A still room is sampled almost for free; a fast reach is frozen with a burst; a settled scene earns one expensive deep look. Spend follows salience — the signal/noise decipherer of the napkin.

The address stamp. Every reading is stamped with normalized coordinates and a timestamp before meaning is attached. Nothing is remembered without its address; x-y-second is the primary key of the whole machine, and any later claim can be walked back to the pixel and the second that produced it.

The macro eye and the helpers. A vision-language model produces the macro picture — “*a macro picture of what’s happening ⇒ VLM*” — which supplies context: what matters in this scene and which specialists to summon. Helpers are then called for the job: a reader for the label, a state extractor for the jar, a listener for the voice. Each is the best available understanding of its one slice of the world; each writes text; none knows the others exist. No helper is architecture — remove any one and the rest carry on.

5.3 TEXT: the only thing kept

The output of extraction is the napkin’s boxed word: TEXT. Text from various helpers, marked by coordinates, appended to an immutable log. Every line carries where, when, which helper, and how sure. This is the entire output of perception: text you could read aloud.

5.4 The binder: physicality and temporal, together

The binder makes many readings into one moment. It does not understand the kitchen; it observes that several lines share a where and a when — “*binding the coordinates, physicality & temporal together*” — and fuses them on that evidence alone. Binding is evidence-driven, not semantic, which is why a wrong binding can be reconciled later (tombstones, Chapter 7) without touching what was read. The output is the napkin’s *first thought*: raw material that can be marinated on. The napkin’s own note is the design’s humility: “*this is not a complete Node, it’s raw material.*”

5.5 Node creation: the second black box

Raw material becomes living cells — “*a physical living breathing cell that can combine with fellow nodes to derive new things — $1+1 = 4$, $\neq 2$.*” A node is a persistent thing (a jar, a person, a shelf) that thickens every time the world confirms it. Nodes combine: co-firing histories derive facts no sensor observed. The box also carries the napkin’s two governance rules: encryption — only the person using the app can decipher (the container, Chapter 12) — and the discarder of meta: derived content that proves useless is deleted; the raw beneath it, never.

5.6 The information supplier: the third black box

A question does not go to the archive; it goes to the supplier, whose contract is the napkin’s phrase *just enough context*. The supplier walks the nodes, selects the few claims that bear on the question — three memories, not three thousand — and packs them with their citations for the reasoner. The supplier is the product decision; retrieval quality, not model quality, is where answers are won.

5.7 The LLM does what an LLM does

The last box is the least special. An ordinary local model receives the packed context and writes the sentence. It is swappable by design; the memory that feeds it is not. Grounding is enforced as a contract: a clause without supporting evidence is not emitted, and the system abstains rather than guesses.

5.8 What flows between stages

One direction, one format. World→Extraction: transient media, dead in milliseconds. Extraction→TEXT: addressed claims. TEXT→Binder: claims; Binder→Nodes: bound moments (raw material). Nodes→Supplier: evidence with provenance. Supplier→LLM: a small cited packet. LLM→user: a sentence whose every clause walks back to a claim — or a refusal. Nothing flows backward; nothing mutates upstream.

CHAPTER 6

One moment, end to end

The worked example is the napkin’s own scene, the same one the interactive walkthrough renders. Times are illustrative; the mechanics are the shipped ones.

6.1 09:14:07, Tuesday, the kitchen

A person reaches for a jar and says something: “*we’re nearly out of the hazelnut stuff.*” Light and sound. This is the only moment that will ever happen; everything after is about keeping it.

6.2 The same second, inside extraction

The blur gate discards the smeared frames of the reach and keeps one legible frame. The spend dial, which had been idling on a still kitchen, spikes for the motion and settles into one deep look. Three helpers write four lines:

09:14:07	x .44 y .48	label reads "NUSSGOLD"	verbatim 1.00
09:14:07	x .31 y .42	glass jar, $\frac{1}{4}$ full, lid ajar	gist 0.72
09:14:07	whole frame	morning kitchen, a reach	gist 0.61
09:14:09	mic	"...nearly out of the hazelnut stuff"	verbatim 0.93

The background is transcribed too — a kettle click at 09:14:05, footsteps leaving at 09:14:12 — because one day the question will be “did I put the kettle on?”, and the answer will exist. The frame that produced all of this lived roughly a hundred milliseconds. Frames kept: zero.

6.3 The binder, that evening

Same x/y region, same second: the label line, the jar-state line, and the spoken line fuse into one bound moment — *09:14, kitchen: the jar is nearly empty, and it was said aloud.* Raw material; not yet a node.

6.4 The night

The moment is absorbed into cells. The JAR node thickens (seen 7 times since 12 May; $\frac{1}{4}$ full today); the PERSON node records that it was said aloud; the PANTRY shelf carries its 41 sightings. Nodes fire together, and the night shift derives a third thing no sensor observed: *you run out roughly every six weeks* — graded as derived, citing the three readings that earned it. A second candidate — “prefers the 400 g size” — finds no rent-paying evidence and is deleted. The raw beneath is untouched, so a better idea tomorrow gets the same evidence.

6.5 Weeks later: the question

“*Do we need more of that spread?*” The supplier walks the nodes and packs three memories: the $\frac{1}{4}$ -full sighting with its date, the spoken sentence, the derived six-week cycle. The model writes: “*Almost certainly. It was a quarter full on 12 June, you said so yourself — and it’s been about six weeks.*” Each clause carries its citation. Ask instead “which brand does my sister buy?” and the answer is “*I have never observed that*” — the refusal is the feature.

6.6 What the example proves

That the pipeline composes: address-stamped extraction makes binding possible; binding makes nodes possible; nodes make derivation possible; provenance makes the final sentence auditable and the refusal principled. And what it does not prove: that extraction is rich enough on real, messy weeks — which is exactly what the falsification gate (Chapter 13) exists to test.

CHAPTER 7

The domain model

The vocabulary below is the live system's (`tracemem`), not its predecessor's. The unit of memory is the *claim*.

7.1 Claim — the atom

A claim is one line of extracted text with its address and its provenance: what was read, where in the frame (normalized coordinates), when (timestamp), by which helper, and how it is graded. Claims are immutable and append-only (I3). The claim log renders back to a human-readable transcript — the memory can be *read*, in order, line by line.

7.2 Grade — witnessed, then downhill

Grades order trust. A claim read directly off the world (a verbatim label, a transcribed sentence) is *witnessed*; everything built above it — bindings, patterns, derived facts — carries a grade that *degrades with derivation depth*, one of the repository's hard laws. The grade is not a float decoration; it is the input to the refusal gate. Deep derivations must be proportionally better-evidenced to be spoken at answer time.

7.3 Episode — the bound moment

The binder's output: claims fused by co-occurrence in space and time into one event. Episodes are projections (rebuildable from the log), carry the evidence tuple that formed them, and are subject to *reconcile*: when later evidence shows a binding was wrong, a tombstone retires it and a corrected episode replaces it — correction as new information, never as mutation.

7.4 Entity and pattern — the living cells

An entity is a node with a history: the jar, the person, the shelf — each a list of dated confirmations. A pattern is a nightly derivation over entities: the six-week cycle, the morning routine. Patterns are the napkin's $1+1=4$ and they are the most disposable layer in the system: the pruner deletes any that stop paying rent, and the raw claims beneath guarantee they can be re-derived or replaced.

7.5 The ask contract

Answering is a typed pipeline: *cue* (read the question), *retrieve* (walk entities and episodes for bearing claims), *pack* (assemble just-enough context, every item citable), *ground* (emit only clauses supported by the packet; abstain otherwise). The contract's output is either a cited answer or an honest abstention — there is no third shape.

7.6 Constructors, not validators

Throughout, invalid states are unrepresentable rather than checked-for: claims without provenance, confidences outside range, bindings without evidence do not pass construction [19][22]. The domain does not trust its callers, including us.

CHAPTER 8

Software architecture

8.1 The order of authority

tracemem’s constitution file states a precedence unusual enough to quote: the napkin outranks the code; the code outranks the tests; the tests outrank the campaign documents; the campaign documents outrank the specs. Prose lost its authority in this project once already (Chapter 14); the ordering makes sure the running system, not the latest document, is what wins arguments.

8.2 Contract-first modules

Every source file carries a contract header: what it OWNS, its IN→OUT, its TEST, and its spec reference. The module tree maps one-to-one onto the napkin’s boxes — `extraction/`, `text/`, `binder/`, `nodes/`, `container/`, `supplier/`, `llm/` — so a reader can hold the drawing in one hand and the repository in the other. Implemented logic is tested green; unimplemented bodies raise `NotImplementedError`, and their acceptance oracles exist ahead of them as `xfail`-marked tests.

8.3 The test suite as ledger

The test run is the project’s status report, by design: *green* is the regression wall (implemented and guaranteed); *xfail* is the TODO ledger, each entry naming the spec section it awaits; *FAILED* means a real bug that outranks all other work. As of 2026-07-21: 274 green, 1 xfail. The remaining stubs are exactly the hardware/model adapter set — the boundary between the thinking machinery (real) and the senses (fakes), stated plainly.

8.4 Ports, adapters, and fakes

The v0 stack is hexagonal in the classical sense [21][22] — domain, application, ports, adapters — and tracemem keeps the discipline where it matters: every hardware and model dependency (OCR, ASR, camera eyes, the sodium cipher, the local LLM) sits behind a port with a deterministic fake. The crown test exercises the entire pipeline on fakes; making an adapter real changes no domain code. Event-sourcing and disposable projections [23][24] are the storage discipline; the strangler fig [25][40] is the migration discipline (Chapter 14).

8.5 The hard laws

Five laws inherited from the blueprint bind all of it: raw is immutable — no update path exists; pixels never leave the device; one model on the GPU at a time (a thermal and correctness constraint learned the hard way on v0); grade degrades with derivation depth; every capture gap is witnessed — if the senses were dark, the memory must say so rather than paper over it.

8.6 The crown test

One test stands above the suite: a shuffled synthetic week is fed through the whole machine — episodes form, entities accrete, nightly patterns derive, a question is asked, a cited answer comes back, and an unanswerable question is honestly abstained from. It passes. It is the executable form of this document’s Chapters 5–7, and it is what “the thinking machinery exists” means when this document says it.

CHAPTER 9

The perception subsystem

9.1 The cheap stratum and the expensive eye

Perception is two economies. The cheap stratum — detection, OCR, speech, sound events, motion — runs always, because it can. The expensive eye — deep VLM description — runs rarely, because it must. The architecture’s job is to spend the expensive looks exactly where the day’s value is.

9.2 Saliency: the spend dial

The dial moves by itself, many times a second, on dwell, novelty, and stability [8]: a still room earns almost nothing; an entering hand spikes the framerate; a settled new scene earns the one deep look. This is event segmentation (Chapter 2) turned into a scheduler, and it is why all-day capture is affordable at all.

9.3 The budget: no overdraft

Deep looks are governed by a token bucket [28]: a hard hourly budget, refilled steadily, spent by saliency, with no overdraft path. Attention is a budget, not a vibe — the system can always account for why it looked hard at one moment and not another.

9.4 The enrichment ladder

Attention deepens in rungs — glance (a jar), read (the label), state ($\frac{3}{4}$ empty, lid ajar), context (a German promo sleeve, therefore probably a German kitchen) — and each rung emits *only the delta*: what the previous rung did not already know. The same fact is never stored twice; storage grows with new knowledge, not with time spent looking.

9.5 The measured frontier

The first real number exists, and it is encouraging without being sufficient: over 111 real captured frames, mechanically scored against frontier-model ceilings, L1-verbatim recall is **0.974** — a 2.6% gap between what the extraction stack reads and what the best available oracle reads. The audit also produced a diagnosed lesson (a cropping law confirmed on meal scenes: 2/4 versus 0/4 on full frames) — the miss-diagnose-fix loop of I7 operating on perception itself. The honest caveat: L1-verbatim is the *mechanical* tier of the claim-match instrument, not the full score, and 111 frames are frames, not weeks.

CHAPTER 10

Consolidation: the night shift

10.1 The job

Read the day's claim log; bind by co-occurrence; thicken entities; derive patterns; refuse weak links; discard failed meta; touch nothing raw. The napkin's underlined words — *always running in the background, but better nightly* — set the schedule: incremental passes keep the memory current; the night pass, on a charging and thermally free device, does the deep work.

10.2 The anatomy

The consolidation heart (M4) is built as guarded stages: *guardrails* (what may never be derived, e.g. assertions that would outrun their evidence); four *derivation kinds* (the taxonomy of what 1+1 may legitimately make); a *selector* (which co-firing histories deserve the model's attention tonight); a *judge* (does the candidate cite enough to survive — refusal is the default); the *night runner*; the *pruner* (rent collection on old patterns); and *remap* (re-pointing structure when reconcile retires an episode). All of it runs today — against synthetic weeks; its inputs become real when the extraction adapters do.

10.3 Accelerator, not gatekeeper

Consolidation makes answers better and faster; it must never become the condition for answering at all. The supplier can serve from episodes alone — unconsolidated memory is slower and flatter, not absent. A memory that only works after a good night's sleep is a demo, not an organ.

10.4 The devil's advocate, kept in the room

Derivation is where a memory system starts lying to itself, which is why the judge prices skepticism in: every candidate pattern must survive an explicit attempt to refute it from the same evidence, and the system deletes its own failed thoughts without sentiment. The napkin's discarder is not cleanup — it is epistemic hygiene.

CHAPTER 11

Recall: grounded answering with calibrated refusal

11.1 The loop

Cue → retrieve → pack → ground. The question is read for its entities, time hints, and kind; the supplier walks the graph; the packet is assembled small — just enough context, every item citable; and the grounding gate lets a clause through only if the packet supports it. The reasoner is local [29], ordinary, and swappable; on v0 the heuristic fallback refuses everything, which is the correct degenerate behavior for a memory: no reasoner, no guessing.

11.2 Refusal as calibration

Refusal is not a failure mode; it is the mechanism that makes open questions (I6) safe. The formal frame is selective prediction [13]: answer only above a confidence bar, and measure the risk–coverage trade rather than assert it. Attribution is the other half [31]: an answer’s citations are checkable objects, so a wrong answer is a diagnosable event, not a shrug.

11.3 What answering feels like

Short, cited, and honest. *“Almost certainly — ¼ full on 12 June, you said so aloud, and it’s been six weeks.”* Or: *“I have never observed that.”* The product bet is that users forgive a memory that sometimes says “I don’t know” and never forgive one that lies once.

Part III · Reality

CHAPTER 12

The privacy architecture and its threat model

12.1 Structural, not procedural

The privacy claims rest on the two structural invariants: no raw persistence (I1) and text-only egress (I2). What does not exist cannot be stolen, subpoenaed, or leaked; what cannot cross the boundary cannot be exfiltrated by a bug in a feature. This is privacy by design in Cavoukian’s strict sense [5] — embedded in architecture, not policy — and it is the strongest available reading of GDPR’s data-minimisation principle [6]: retain meaning, not medium.

12.2 The container

Retained text lives in the `.trace` container: a specified format anyone can implement and only the user can open. Three layers with deliberately asymmetric guarantees: a header readable without the key (version, device, counts); the RAW claim log — append-only, immutable, encrypted; and the META layer — nodes, edges, confidence — encrypted, *deletable*, and *re-derivable*. Lose the meta and nothing is lost; lose the raw and everything is. Encryption at rest is AES-256-GCM [33] on v0; `tracemem`’s cipher adapter (libsodium) is presently stubbed — recorded as such in the ledger on page 3, because a privacy document that overstates its own crypto status has already failed.

12.3 Threats, walked

Device theft: the container is ciphertext under the user’s key; the header leaks counts, not content. *Cloud breach*: there is no cloud; nothing to breach. *Subpoena*: what can be produced is the text the user can already read — and nothing else exists. *Malicious app update*: the egress guard is the choke point; media has no path out to leak. *A compromised model vendor*: models run locally; the reasoner sees packed text, and only on device. The honest residual in each case is the text itself — which is why the last section of this chapter exists.

12.4 What architecture cannot discharge

Honesty requires the residue stated. Perceiving people into text still touches privacy: “Anna said she’s pregnant” is sensitive with no pixel involved. The bystander literature on wearable cameras [34] transfers in part to any wearable perception. Product-level obligations therefore remain: disclosure norms for wearers, redaction policies for bystander speech, retention and export controls in the user’s hands, and the wearer’s own social contract. The architecture makes the worst artifact impossible; it does not make the remaining ones weightless. This section exists so that no investor, customer, or regulator can say the project hid the residue behind the moat.

CHAPTER 13

The evaluation instrument

13.1 The philosophy

Two failure modes destroy memory products: fabrication, and silent information loss at capture time. The instrument is built to expose both, mechanically, so that every miss becomes a work item rather than a debate.

13.2 Scoring answers

The v0 instrument scores a question battery with fabrication priced in: a made-up answer costs what a correct one earns, so the headline number cannot be inflated by confident guessing. Every miss is mechanically diagnosed as (A) never captured or (B) captured but unused — the (A)/(B) split is the steering wheel: (A) misses buy sensors; (B) misses buy binder and reasoner work. The v0 baseline on its 92-second proving clip was measured at RAS 40.0 with 27.3% hallucination, 8 of 9 misses diagnosed (B) — the honest starting line that motivated the greenfield rebuild’s discipline, recorded here because a thesis that hides its worst number teaches its readers to distrust its best one.

13.3 Ceilings: knowing the maximum before claiming the score

tracemem’s harness pins *ceilings*: frontier-model readings of the same frames, cached and versioned, so every extraction score is a fraction of a known maximum rather than a free-floating number. Current audited state: 111 ceiling frames carrying 4,674 claims; against their mechanical tier the extraction stack reads 97.4% (Chapter 9). Ceilings are re-pinned when the frontier moves; the score is always *relative to the best reader we can rent*, which is the only honest definition of “lossless-enough.”

13.4 Rehearsal, and the verdict format

Before real weeks, synthetic ones: a generated week with known ground truth is pushed through the entire machine, and the report grades strata separately — verbatim recall, factual recall, emergent (derived) recall — with a one-word verdict. The first full rehearsal returned its verdict honestly: `FIX_UPSTREAM` — verbatim 1.0, factual 0.5, emergent 0.0, with the gap diagnosed to specific upstream stages and two invalid questions flagged. The system grading itself harshly, in machine-readable form, is the culture the instrument exists to enforce.

13.5 The falsification gate

The gate that matters is still ahead, and this document refuses to blur that: capture real weeks on the real device, run the full battery, diagnose every miss. The hypothesis (Chapter 1) lives or dies there. Everything in Part II exists to make reaching that gate a matter of adapters, not architecture.

CHAPTER 14

Two stacks, one product: an honest history

14.1 v0: the stack that runs

The reader doing diligence will find two repositories, and deserves the story before the surprise. v0 has run continuously since June 2026: live capture feeding an always-on hub, a nightly consolidation job at 03:30, an encrypted store, an iPhone capture app, and an evaluation battery. It produced the first disciplined binder (36 entities, 58 bindings, 144 weak links refused on the proving clip), the first honest scores, and most of the hard laws — one model on the GPU, capture goes dark silently if crowded, the embedding endpoint that had to be replaced. It also accumulated what every v0 accumulates: prose documents that drifted from the code until, on 2026-07-17, the founder deleted every one of them and declared that the present tense lives only in code, tests, and running processes. That deletion is why this document cites ledgers instead of promises.

14.2 The greenfield: the napkin, rebuilt

Two days later the architecture was redrawn by hand — the napkin — and `tracemem` was scaffolded from it: specs first, contract headers in every file, fakes behind every port, the crown test as the spine, `xfail` as the TODO ledger. The napkin was transcribed verbatim into the repository as its constitution, outranking everything including the specs. v0's lessons are inherited as laws rather than as code.

14.3 The strangling sequence

The migration follows the strangler fig [25][40]: v0 keeps running as the capture organ and the measuring baseline while `tracemem`'s adapters come real in dependency order — cipher, OCR/ASR, the camera eyes, the phone tap. Each adapter that lands moves one organ from the old body to the new; the crown test and the ceilings verify equivalence at each step. v0 is decommissioned by explicit founder decision when the falsification gate runs green on the new stack — not by drift, and not before.

14.4 What v0 knows that a rewrite would forget

The operational scar tissue is the inheritance: that capture dies silently when the GPU is shared, and must be watched by liveness checks; that embeddings needed a different engine than the reasoner; that phone deploys stall in reproducible ways; that a store must be snapshotted and shadow-served to measure it without killing capture. None of this is in any paper; all of it is in the laws and runbooks the new stack starts with.

CHAPTER 15

Honest risks

The hypothesis may be false. Text may lose something questions need at a rate the enrichment ladder cannot close. This is the existential risk; it is measurable (Chapter 13), the gap today is 2.6% on frames, and the number on real weeks is unknown. If the gap on lived weeks proves irreducible, the architecture does not survive by pivoting to stored media — the project fails honestly instead.

The senses are not real yet. The thinking machinery runs; the hardware adapters are stubs. The risk is not that adapters are hard — v0 proves each exists — but that their *quality* on device, on battery, all day, is below what the ceilings assume. The battery and thermal budget on glasses-class hardware is unmeasured and is called out as such.

A single founder built this. Bus factor one, and the seat this document exists to fill is empty (Chapter 17). The mitigations are the ones in evidence: specs with contracts, a constitution file, tests as ledger — the repository is built to be joined.

Two stacks is a tax. Every week both run is a week of split attention. The strangling plan bounds it, but the tax is real and admitted.

Local models may plateau. If on-device reasoning stalls, I2 permits routing *text* to a stronger model — a privacy-preserving escape hatch that is nonetheless a product compromise and is treated as one.

The social residue. Chapter 12's residue section is a risk, not a footnote: a wearer whose glasses transcribe speech is a social object the world has already rejected once. The bet is that text-only, cited, user-auditable memory is on the acceptable side of the line the recorders died on. That bet can lose.

CHAPTER 16

Position and moat

16.1 The empty quadrant

Physical-world context with zero retention: no shipping product occupies it. The recorders that tried the adjacent quadrant died there — Humane’s pin shut down with its assets sold to HP for \$116M (February 2025); the always-on recorder category is a graveyard because storage-first capture meets the wall of Chapter 1. The digital-only assistants — however personalized — cannot see the room. The quadrant is empty because physical context has always required retention. TRACE’s architecture is the first way in without it.

16.2 Why now

Three curves cross. On-device AI crossed the capability threshold — billions of phones now run local models, and sub-second on-device vision is commodity. AI wearables ship in volume — roughly seven million camera glasses in 2025 alone — proving demand for the form factor while their retention architecture proves the objection. And the EU AI Act’s biometric provisions (in force since February 2025) turn “structurally non-retaining” from a philosophy into a procurement criterion, on the continent where this company is being built.

16.3 Why incumbents don’t simply build it

Their business models monetize retained data and engagement; their privacy stacks deliberately stop at the digital life. A zero-retention physical layer contradicts the data economics that fund them. If the capability matters to them, they license it or acquire it — both outcomes require TRACE to exist first.

16.4 The moat, precisely

Not the model (commodity, by design), not the app. The moat is the posture — input-only, provable, GDPR-native — which storage-first competitors cannot copy without abandoning their foundation; the evidence corpus — every deployed week compounds the question batteries and diagnosis logs that tune extraction; and the licensing position: a specified container and a text-only context layer that any device maker can adopt, on the precedent of Arm and Dolby — companies that own a layer, not a box.

CHAPTER 17

The road, and the ask

17.1 The road

Now → 6 months — harden. Make the extraction adapters real (cipher, OCR, ASR, eyes); run the falsification gate on real captured weeks; close the diagnosed gaps until the battery scores survive contact with a lived life. *6 → 12 months — the EXIST phase.* Wearable form factor off the phone; closed beta; the founding team complete. *12 → 24 months — ship.* Public product, multi-device, first licensing conversations from an evidence position.

17.2 The open seat

This document is, among other things, an offer: a technical cofounder seat, equity not employment. The profile is deliberately open across the six readers of the reading guide — the work spans perception, systems, and product — but the seat owns one thing above all: making the senses real and proving the hypothesis on real weeks, with the founder, at parity.

17.3 The first 90 days

Day one is not a whiteboard: it is a repository with a constitution, 274 green tests, one xfail ledger, and a falsification gate waiting for real weeks. The first 90 days are the strangling sequence's first organs — cipher adapter, OCR/ASR, the first real captured week through the crown pipeline — and the first gate run whose verdict neither of us can predict. That verdict is the point. This is a project that has arranged to be told it is wrong.

17.4 What would change our minds

If the falsification gate shows an irreducible extraction gap on questions that matter, the hypothesis dies and the project with it — honestly. If refusal rates make the product unusable at the confidence floor the ethics require, the bet on I5/I6 was wrong. If a year of evidence shows the social residue is rejected even without retention, the market is not there. Each of these is written down *before* the evidence arrives, which is the only time such sentences can be written credibly.

Appendix A — The napkin, verbatim

Handwritten by the founder; photographed 2026-07-18; transcribed into the repository as its constitution. On any conflict it outranks code, tests, campaign documents, and specs — in that order below it. Nothing in it may be “improved” in place; a change to the architecture is a founder decision, recorded on the board.

Standard Model

```
[World] --Senses--> Eyes / Ears / Organs --> [Brain]
      ^
      |
(because we can't manufacture other senses
or most of the information is obtained through this)
```

EXECUTION

```
[World] -> Video (Eyes), Audio (Ears)
  -> [Extraction Black Box]
  -> Coordinates + time stamps
  -> Blurry frames discarded · Signal-Noise decipherer + Context
  -> A macro picture of what's happening => VLM
  -> A specialized eye or helper for various tasks are called (Helper)
  -> Best understanding possible of the world
  -> [TEXT]
```

Text from various helpers marked by coordinates

```
-> [Binder] => binding the coordinates, physicality & temporal, together
-> First thought or raw material that can be marinated on
    Note: This is not a complete Node, it's raw material
-> [Node Creation Black Box]
    · Encryption, i.e. people using our app can decipher
    · A physical living breathing cell that can combine with
      fellow nodes to derive new things —  $1+1 = 4$ ,  $\neq 2$ 
    · Always running in the background but better nightly
    · Discarder of meta, i.e. the marinated content that's not
      useful is discarded, NOT the raw material
```

```
[Information Supplier Black Box]
  -> Just Enough Context for local LLM
  -> LLM does what LLM does
```

The box→code map (subordinate to the drawing):

Napkin box	Code
Extraction Black Box	src/tracemem/extraction/
TEXT (raw, immutable, coordinate-marked)	src/tracemem/text/
Binder	src/tracemem/binder/
Node Creation Black Box	src/tracemem/nodes/ · src/tracemem/container/

Napkin box	Code
Information Supplier Black Box	src/tracemem/supplier/
LLM does what LLM does	src/tracemem/llm/

Appendix B — Glossary

claim	one line of extracted text with address, provenance, and grade — the atom of memory
grade	ordered trust level; <i>witnessed</i> at the top, degrading with derivation depth
episode	claims bound by shared place and second — the napkin’s “first thought,” raw material
entity	a persistent node with a dated history (the jar, the shelf, the person)
pattern	a nightly derivation over entities — the napkin’s $1+1=4$; deletable, re-derivable
tombstone	the record that retires a wrong binding without mutating the log
ceiling	a frontier-model reading of the same frame, pinned and cached — the denominator of every extraction score
crown test	the end-to-end test: synthetic week → episodes → entities → patterns → cited answer → honest abstention
supplier	the third black box: cue → retrieve → pack → ground; three memories, not three thousand
.trace	the container: readable header · encrypted immutable RAW · encrypted re-derivable META
(A)/(B) miss	diagnosis of a failed answer: never captured (A) versus captured but unused (B)
strangler fig	the migration pattern: v0 keeps running while <code>tracemem</code> ’s adapters replace its organs

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